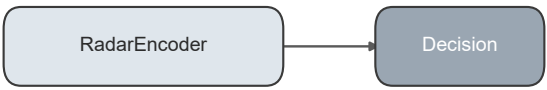


(a) Four architectures compared

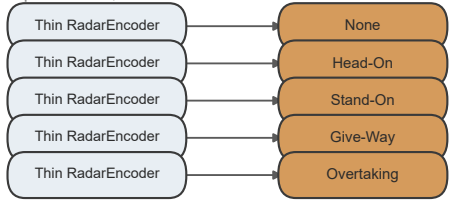
Single network

369K · one perception, one decision



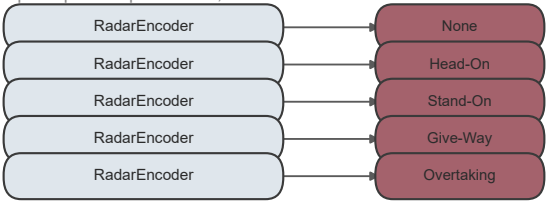
Separate experts, thin

363K · perception duplicated x5, each narrowed



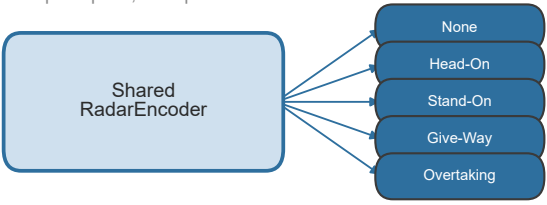
Separate experts, full

1.83M · perception duplicated x5, full width

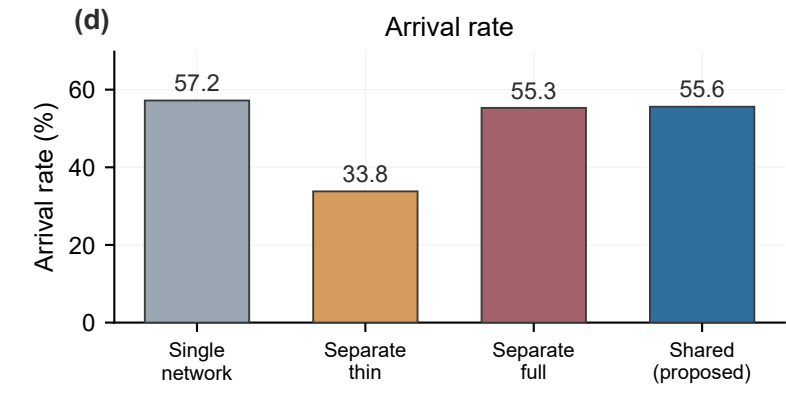
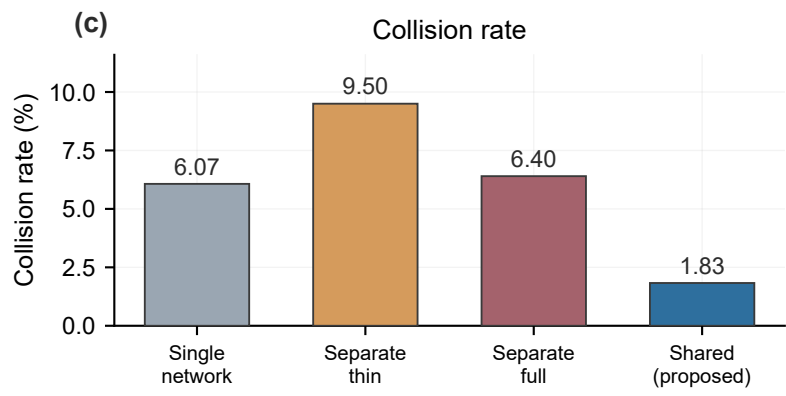
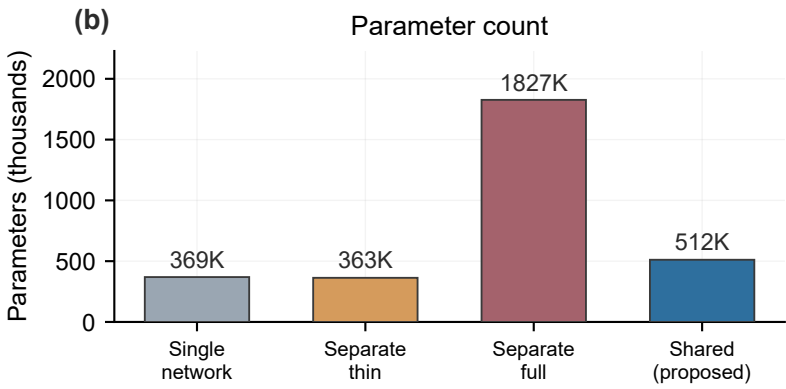


Shared perception (proposed)

512K · one perception, five specialised decisions



Radar perception is ~89% of single-network parameters.
The same expert decomposition is applied to MessageActor, ControlActor and Critic.



Separate-thin is a single seed (no error bar shown; not used for effect-size claims). Single-network arrival includes one run that converged to a rushing policy (12.5% collision); excluding it gives 53.8%. Separate-full metrics come from the same completed runs used in Fig.4 (dim 6) and Fig.5. The physical scene is a shared perception problem, whereas COLREGs obligations create situation-specific decision problems.